

CCNA 200-301 v1.1	Switching
Basic Switch Commands	
enable	Move into privileged mode
configure terminal	Move into global configuration mode
hostname [name]	Give the switch a name
enable secret [password]	Require a password to enter privileged mode
interface [type][number] interface g0/7	Move into interface configuration mode
interface range [type] [port-range] Interface range tengigabit 0/1-4	Select range of interfaces to apply configurations to as a group
exit	Exit out of the current mode to the lower mode
end	Jump back to the “top” of privileged mode
shutdown	Administratively shut down the interface
no shutdown	Administratively bring up the interface
copy running-config startup-config	Save the configuration
Switch Show Commands	
show version	Display information about the device
show interfaces	Display information about all interfaces on the device
show interfaces [type][number] show interfaces g0/7	Display information about a particular interface
show interfaces status	Show table of interfaces, status, VLAN, duplex/speed, interface type
show ip interfaces brief	Displays a table of interfaces, their IP addresses, up/down status
show running-config	Displays the current running configuration
show startup-config	Displays the configuration that will load upon reboot
show cdp neighbors	Display basic info about immediately connected CDP devices
show cdp neighbors detail	Detailed information about CDP neighbors, including their IP address
MAC Table Commands	
show mac address-table	Show the entire MAC address table, all addresses, VLANs and ports
show mac address-table dynamic	Show dynamically learned MAC addresses
show mac address-table static	Show statically configured MAC addresses
show mac address-table interface <type><#> show mac address-table interface g0/10	Show MAC addresses on a specific interface
show mac address-table vlan <vlan id> show mac address-table vlan 1	Display the MAC addresses associated with a specific VLAN
clear mac address-table dynamic	Clear the dynamically learned MAC addresses from the MAC table
[no] mac address-table static <MAC address> vlan <vlan id> interface <type><#> mac address-table static aaaa.bbbb.cccc vlan 10 interface f0/10	Manually add/remove a static MAC to the MAC address table Bind it to a specific VLAN and interface
show mac address-table aging-time	Display aging time (in seconds) for dynamically learned address
mac address-table aging-time <seconds> vlan <id> mac address-table aging-time 300 vlan 10	Configure aging time for dynamic MAC addresses on a specific VLAN
show mac address-table address <MAC address> show mac address-table address aaaa.bbbb.cccc	Display VLAN and interface associated with a specific MAC

Switching Mode Commands	
switching-mode fragment-free	Validate the first 64 bytes of the frame as it comes in
switching-mode store-and-forward	Buffer the entire frame before forwarding
switching-mode cut-through	Do not wait to buffer the frame; start forwarding immediately
STP Port Configuration Commands	
spanning-tree	Enable spanning-tree on the switch (enabled by default)
show spanning-tree	Display STP information for the switch
show spanning-tree vlan <vlan-id>	If you want to see STP details for a specific VLAN
show spanning-tree vlan 100	
spanning-tree mode <pvst rapid-pvst mst>	Change STP to one of its variant modes
spanning-tree mode rapid-pvst	rapid-pvst (Rapid PVST+) is the preferred these days
spanning-tree vlan <vlan-id> priority <value>	Influence the root bridge election process
spanning-tree vlan 20 priority 4096	Priority can be 0 – 61440 in increments of 4096 Lowest is most likely to win
spanning-tree portfast	Put the switchport into portfast mode
spanning-tree portfast default	Enable portfast on all ports globally
STP Port Configuration Commands	
spanning-tree bpduguard enable	Prevent a PortFast-enabled port from sending or receiving BPDUs – no guardrails against a loop
spanning-tree portfast bpduguard default	Enable BPDUGuard globally on all access ports
spanning-tree bpduguard enable	Disable a specific PortFast-enabled port if it receives a BPDU – put port in err-disable state
spanning-tree portfast bpduguard default	Enable BPDUGuard globally for all access ports
spanning-tree vlan <vlan-id> root primary	Manually set a switch to be the root bridge
spanning-tree vlan 3 root primary	
spanning-tree vlan <vlan-id> root secondary	Manually set a switch to be a secondary root
spanning-tree vlan 3 root secondary	No strict limit – but best to only have one
Spanning Tree Timer Commands	
spanning-tree vlan <vlan-id> hello-time <seconds>	Adjust hello-time timers
spanning-tree vlan 50 hello-time 3	From 1 – 10 seconds; Default is 2 seconds
spanning-tree vlan 50 forward-time <seconds>	How long a port stays in the listening and learning states before transitioning to the forwarding state From 4 – 30 seconds, split evenly between listening and learning states
spanning-tree vlan 50 max-age <seconds>	Set how long a switch will wait before aging out BPDU info if no new BPDU is received; From 6 – 40 seconds; Default is 20
Spanning Tree Show Commands	
show spanning-tree	Provide details about the root bridge, local bridge, and port roles for all VLANs
show spanning-tree summary	Display summary of spanning tree status across all VLANs
show spanning-tree vlan <vlan-id>	Display Spanning Tree information for a specific VLAN
show spanning-tree vlan 1	
show spanning-tree interface <interface-id>	Display Spanning Tree information for a specific interface
show spanning-tree interface fa0/9	

Rapid PVST+ Commands	
spanning-tree mode rapid-pvst	Enable Rapid PVST+ on the switch
show spanning-tree	Display configuration, including mode (Rapid PVST+), root bridge for each VLAN, and port roles
show spanning-tree vlan <vlan-id> show spanning-tree vlan 15	Shows STP details for a specific VLAN, including the root bridge, timers, and port roles
spanning-tree portfast	Immediately transition the port to forwarding state
spanning-tree portfast default	Apply PortFast to all access ports
Rapid PVST+ Port Configuration Commands	
spanning-tree bpdupfilter enable	Prevent a PortFast-enabled port from sending or receiving BPDUs – at risk of loops
spanning-tree portfast bpdupfilter default	Enable it globally for all access ports
spanning-tree bpduguard enable	Prevent loops by disabling a PortFast-enabled port if it receives a BPDU – put port in blocking err-disable state
spanning-tree portfast bpduguard default	Enable it globally for all access ports
spanning-tree vlan <vlan-id> priority <value>	Lowers switch priority for a specific VLAN
spanning-tree vlan 3 priority 4096	Increase its chances of becoming the root Range from 0 (highest priority) to 61440 (lowest) in 4096 increments
spanning-tree vlan <vlan-id> root primary spanning-tree vlan 3 root primary	Manually set a switch to be the root bridge
spanning-tree vlan <vlan-id> root secondary spanning-tree vlan 3 root secondary	Manually set a switch to be a secondary root
Rapid PVST+ Show Commands	
show spanning-tree	Display detailed information about the root bridge, local bridge, and the state of each port in the spanning tree topology
show spanning-tree interface <interface-id> show spanning-tree interface fa0/10	Display detailed STP information for a specific interface, including its current state (Forwarding, Learning, etc.), role, and cost
show spanning-tree summary	Provides a summary of the STP configuration, including BPDU Guard, PortFast, and other global setting

Rapid PVST+ Timer Commands	
spanning-tree vlan <vlan-id> hello-time <seconds> spanning-tree vlan 1 hello-time 1	Adjust hello-time timers 1 – 10 seconds; Default 2
spanning-tree vlan <vlan-id> forward-time <seconds> spanning-tree vlan 1 forward-time 10	Adjust forward-delay timers 4 – 30 seconds, Default 15
spanning-tree vlan <vlan-id> max-age <seconds> spanning-tree vlan 1 max-age 15	Adjust max-age timers 6 – 40 seconds, Default 20
VLAN Commands	
vlan [vlan-id] vlan 10	Create a VLAN
no vlan [vlan-id]	Delete a VLAN from the VLAN database
name [vlan-name] name Marketing	Name a VLAN
switchport mode access	Turn a switchport into an access port
switchport access vlan [vlan-id] switchport access vlan 10	Assign a VLAN to an access port
VLAN Show Commands	
show vlan	Display VLAN database
show vlan brief	Display VLAN information
show interfaces [interface-id] switchport show interfaces FastEthernet 0/1 switchport	Check VLAN assignment to interfaces
show vlan id [vlan-id] show vlan id 10	Check VLAN membership on a specific port
Trunk Link Negotiation Commands	
switchport mode access	Set the port to be a Layer 2 access port only
switchport trunk encapsulation dot1q	Necessary on some switches before issuing the switchport mode trunk command
switchport mode trunk	Set the port to be a trunk port
	Actively negotiate to form a trunk link
switchport nonegotiate	Set the port to not negotiate a trunk link
	Assumes the trunk will be manually configured
switchport mode dynamic desirable	Set the port to actively negotiate a trunk link The trunk link forms if negotiation is successful
switchport mode dynamic auto	Set the port to not negotiate, but be willing to become a trunk link if the other side negotiates
switchport trunk native vlan [vlan-id] show interfaces trunk	Change the native VLAN on an interface Display any trunk links and their status
VLAN Pruning Commands	
switchport trunk allowed vlan <id, id,...>	List the allowed VLANs
switchport trunk allowed vlan 10,20,30	All others are pruned out
switchport trunk allowed vlan remove <id> switchport trunk allowed vlan remove 30	Prune out an existing allowed VLAN
switchport trunk allowed vlan add <id> switchport trunk allowed vlan add 40	Add a specific VLAN to the allow list

VTP Commands	
vtp domain <domain-name> vtp domain company.com	Set the VTP domain name
vtp mode {server client transparent off} vtp mode transparent	Set the switch's VTP role Default is server
vtp password <password> vtp password pass1234	Configure a password for VTP messages in a VTP domain All switches in a VTP domain must have the same password
VLAN Interface Commands	
ip routing	(If necessary) enable IP routing on the switch
interface vlan [vlan ID] interface vlan 10	Enter VLAN interface mode
ip address [address] [subnet mask] ip address 192.168.10.1 255.255.255.0	Assign an IP address to the VLAN interface
no shutdown	Administratively turn the port on
show ip interface brief	Check the VLAN int status
show ip route	Show the routing table Confirm that the switch is ready to route traffic between VLANs
Phone and PC VLAN Commands	
switchport access vlan [data-vlan-id] switchport access vlan 5	Assign the data VLAN to the switchport
switchport voice vlan [voice-vlan-id] switchport voice vlan 7	Add the voice VLAN to the same switchport
mls qos trust cos	Configure QoS to give voice traffic priority
spanning-tree portfast	Set the port to portfast; no delay when using the phone
PoE Commands	
power inline auto	Automatically detect and supply power to connected devices that are PoE-capable (e.g. IP phones, WAPs, security cameras)
power inline never	Disable PoE on a specific interface
power inline static max [max-milliwatts] power inline static max 30000	Configure a maximum amount of power (in milliwatts) that a specific port can supply to connected devices
power inline priority [critical high low] power inline priority critical	Set priority for that port (connected device) in case the PoE budget becomes strained and the switch has to make decisions about which devices to power
Port Security Commands	
switchport port-security	Enable port security on a specific interface
switchport port-security maximum [number] switchport port-security maximum 2	Set the maximum number of allowed MAC addresses on a port
switchport port-security mac-address [mac-address] switchport port-security mac-address 0001.2345.6789	Manually specify the MAC address that is allowed in the interface
switchport port-security mac-address sticky	Save the first MAC address heard on the port as the authorized MAC
switchport port-security violation [protect restrict shutdown] switchport port-security violation protect	Specify the violation response Default is shutdown

MAC Aging Commands	
mac address-table aging-time [time] mac address-table aging-time 600	Configure MAC aging time globally for all VLANs
mac address-table aging-time [time in seconds] vlan [vlan-id] mac address-table aging-time 300 vlan 10	Configure MAC aging time for a specific VLAN
show mac address-table aging-time	Verify MAC aging time
Data and Voice Port Security Commands	
switchport voice vlan [vlan ID] switchport voice vlan 7	Configure the switch access port with the voice VLAN ID
switchport port-security maximum [number] switchport port-security maximum 2	Limit the number of allowed MAC addresses on the switchport
switchport port-security violation [protect restrict shutdown] switchport port-security violation protect	Set the violation response on the switchport
Routed Port Commands	
interface <type><#> interface g0/1	Enter interface configuration mode
no switchport	Convert a switchport from Layer 2 to Layer 3
ip address [address] [subnet mask] ip address 10.1.1.1 255.255.255.0	Assign an IP address and subnet mask
no shutdown	Turn the port on
show ip route	View the route table SVIs and Routed ports will show as “directly connected”
EtherChannel Commands	
channel-group [number] mode on channel-group 1 mode on	Configure static EtherChannel
channel-group [number] mode desirable channel-group 1 mode desirable	Configure physical ports to actively negotiate creating an EtherChannel using PAgp
channel-group [number] mode auto channel-group 1 mode auto	Configure physical ports to not actively negotiate, but be willing to form an EtherChannel using Pagp if the other side is set to mode desirable
channel-group [number] mode active channel-group 1 mode active	Configure physical ports to actively negotiate creating an EtherChannel using LACP
channel-group [number] mode passive channel-group 1 mode passive	Configure physical ports to not actively negotiate, but be willing to form an EtherChannel using LACP if the other side is set to mode active
EtherChannel Show Commands	
show etherchannel	Check the group state
show etherchannel summary	Show the various channel groups and member ports
show lacp neighbor	Display information about the other side of the LACP link
show pagp neighbor	Display information about the other side of the PAgp link
show etherchannel port-channel	Check port-channel status
show interfaces port-channel	Display information on all port-channels